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FRACTIONS

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🔢 What is a Fraction?

A **fraction** represents a part of a whole.

Example:

$\frac{3}{4}$ means 3 parts out of 4 equal parts.

- **Numerator**: The top number (shows how many parts are taken)
- **Denominator:** The bottom number (shows total equal parts)

🧮 Types of Fractions

Type	Example	Description
Proper Fraction	$\frac{3}{5}$	Numerator < Denominator
Improper Fraction	$\frac{9}{4}$	Numerator > Denominator
Mixed Number	$2\frac{1}{3}$	Whole number + proper fraction
Like Fractions	$\frac{1}{4}, \frac{3}{4}$	Same denominator
Unlike Fractions	$\frac{1}{3}, \frac{2}{5}$	Different denominators
Equivalent Fractions	$\frac{2}{4} = \frac{1}{2}$	Same value, different form

➕ 1. Addition of Fractions

✅ Like Denominators:

- Add numerators, keep denominator same

$\frac{2}{7} + \frac{3}{7} = \frac{5}{7}$

✅ Unlike Denominators:

- Find LCM, convert to like fractions, then add

$\frac{1}{2} + \frac{1}{3} = \frac{3}{6} + \frac{2}{6} = \frac{5}{6}$

➖ 2. Subtraction of Fractions

- Same method as addition

$\frac{5}{6} - \frac{1}{2} = \frac{5}{6} - \frac{3}{6} = \frac{2}{6} = \frac{1}{3}$

✖️ 3. Multiplication of Fractions

- Multiply numerators together and denominators together

$\frac{2}{3} \times \frac{4}{5} = \frac{8}{15}$

✅ For mixed numbers:

Convert to improper first:

$1\frac{1}{2} \times \frac{2}{3} = \frac{3}{2} \times \frac{2}{3} = 1$

➗ 4. Division of Fractions

- Keep the first fraction
- Change ÷ to ×
- Flip (reciprocal) the second fraction

$$\frac{3}{4} \div \frac{2}{5} = \frac{3}{4} \times \frac{5}{2} = \frac{15}{8} = 1\frac{7}{8}$$

🔁 Converting Mixed Numbers

- To Improper:
  $2\frac{1}{3} = \frac{(2\times3)+1}{3} = \frac{7}{3}$
- To Mixed:
  $\frac{10}{3} = 3\frac{1}{3}$

🧠 Tips to Remember

- Always simplify your final answer.
- Use LCM for addition/subtraction of unlike fractions.
- Change mixed numbers to improper before multiplying/dividing.
- Reciprocal is needed only in division.